

This In Class Activity is due at the end of class today.

Student 1 name and ODIN id: _____

1. Write down the names of the **5 ANSI joins** and what each does.

2. What are the rules for using **UNION**, **INTERSECT**, and **EXCEPT**?

3. **Set Up the Tables in DBeaver:**

*Type the SQL statements below in DBeaver to **practice typing queries manually instead of copy pasting** and get more comfortable with SQL syntax. This will help you build muscle memory and understand table structure better.*

```
CREATE TABLE students (  
  student_id INT,  
  name VARCHAR(50),  
  major VARCHAR(50)  
);  
INSERT INTO students VALUES  
(1, 'Alice', 'CS'),  
(2, 'Bob', 'Math'),  
(3, 'Carol', 'Physics'),  
(4, 'Dave', 'CS');  
CREATE TABLE club_members (  
  student_id INT,  
  club_name VARCHAR(50)
```

```
);  
INSERT INTO club_members VALUES  
(1, 'Robotics'),  
(2, 'Mathletes'),  
(5, 'Chess Club');
```

4. For these parts, type your sql in Dbeaver, check your answers and if you think that's correct write it down here.

- List names and clubs for students who are in a club.
- List *all* students, whether or not they're in a club.
- List students who are **not** in any club.
- List all students and club members (even if not matched).
- Show every possible combination of students and clubs.

5. Imagine this dataset is part of a campus app.

- What other entities would you add? *Write 2 more potential entities.*

- What extra attributes would you add to those entities?
6. Translate the given English language query to SQL and run the SQL with your **Spy database**. Show your SQL text, indicate the number of rows returned by the query and provide a screen capture of the first 5 rows returned by the query. If the query returns fewer than 5 rows then list all of the rows returned.
- Find the team name for all teams with at least one agent who speaks Cherokee.
7. Write a query to find the number of agents who speak each language. List the languages in descending order of the number of agents who speak that language.
- List each mission along with its security clearance description
8. Write a query to find the number of agents who speak each language. List the languages in descending order of the number of agents who speak that language.
- List the security clearance description for all missions, and include security clearances that are not assigned to any missions.

